23/02/23

es 1

$$y(t) = e(t) + \frac{1}{2} e(t-1) - e(t-2) \qquad e(t) \sim WN(0,1) \rightarrow MA(2)$$

$$\xrightarrow{e(t)} \boxed{W(z)} \xrightarrow{y(t)} \qquad W(z)?$$

$\rightarrow$ operator delay $z^{-1} \Rightarrow z^{-K} v(t) = v(t-K)$

$$\Rightarrow y(t) = e(t) + \frac{1}{2} z^{-1} e(t) - z^{-2} e(t) \Rightarrow W(z) = \frac{y(t)}{e(t)} = 1 + \frac{1}{2} z^{-1} - z^{-2}$$
$$= \left[1 + \frac{1}{2} z^{-1} - z^{-2}\right] e(t)$$

$$\mathbb{E}[y(t)] = \mathbb{E}\left[e(t) + \frac{1}{2} e(t-1) - e(t-2)\right] = \dots = 0$$

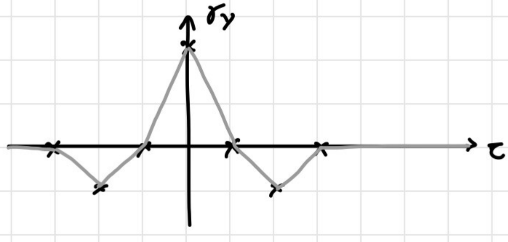
$$\gamma_y(t,0) = \mathbb{E}[y(t)^2] = \mathbb{E}[e(t)^2] + \frac{1}{4}[e^2(t-1)] + \mathbb{E}[e(t-2)]$$
$$+ \cancel{\mathbb{E}[e(t)e(t-1)]}^{0} - 2\,\cancel{\mathbb{E}[e(t)e(t-2)]}^{0} - \cancel{\mathbb{E}[e(t-1)e(t-2)]}^{0}$$
$$= 1 + \frac{1}{4} + 1 = \frac{9}{4}$$

$$\gamma_y(t,1) = \mathbb{E}[y(t)y(t-1)] = \mathbb{E}\left[\left(e(t) + \frac{1}{2} e(t-1) - e(t-2)\right)\left(e(t-1) + \frac{1}{2} e(t-2) - e(t-3)\right)\right]$$
$$= \frac{1}{2}\mathbb{E}[e(t-1)] - \frac{1}{2}\mathbb{E}[e(t-2)] = \frac{1}{2} - \frac{1}{2} = 0$$

$$\gamma_y(t,2) = \mathbb{E}[y(t)y(t-2)] = \mathbb{E}\left[\left(e(t) + \frac{1}{2} e(t-1) - e(t-2)\right)\left(e(t-2) + \frac{1}{2} e(t-3) - e(t-4)\right)\right]$$
$$= -\mathbb{E}[e(t-1)] = -1$$

$$\gamma_y(t,3) = \mathbb{E}[y(t)y(t-3)] = \mathbb{E}\left[\left(e(t) + \frac{1}{2} e(t-1) - e(t-2)\right)\left(e(t-3) + \frac{1}{2} e(t-4) - e(t-5)\right)\right]$$
$$= 0$$

$$\gamma_y(t,\tau) = 0 \quad \tau \geq 3$$



es 2

$$y(t) = \underbrace{\tfrac{1}{2}y(t-1) - \tfrac{1}{4}y(t-2)}_{\text{auto regressive process AR(2)}} + e(t) \qquad e(t) \sim WN(0,1)$$

operator delay $\Rightarrow$ $y(t) = \frac{1}{2}z^{-1}y(t) - \frac{1}{4}z^{-2}y(t) + e(t)$

$\rightarrow$ $e(t) = (1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2})y(t)$

$$\Rightarrow y(t) = \frac{1}{\underbrace{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}_{\text{transfer function}}}\, e(t)$$

$\mathbb{E}[y(t)] = \mathbb{E}[\frac{1}{2}y(t-1) - \frac{1}{4}y(t-2) + e(t)] =$

$= \frac{1}{2}\mathbb{E}[y(t-1)] - \frac{1}{4}\mathbb{E}[y(t-2)] + \cancel{\mathbb{E}[e(t)]}^{0} =$ Hyp: $y(t)$ is SSP

$= \frac{1}{2}m - \frac{1}{4}m$

$\rightarrow m = \frac{1}{2}m - \frac{1}{4}m \Rightarrow m = 0$

THEOREM – If $y(t) = W(z)v(t)$ and $v(t)$ is a SSP, $W(z)$ is asymptotically stable then $y(t)$ is stationary

$W(z)$ is as. stable ? $\quad W(z) = z^2 / z^2(1 - \frac{1}{2}z^{-1} - \frac{1}{4}z^2)$

$= z^2 / z^2 + \frac{1}{2}z - \frac{1}{4}$

↳ poles need to be strictly the complex unit circle

if also the zeroes are in the circle it is minimum-phase

$z^2 + \frac{1}{2}z + \frac{1}{4} = 0 \rightarrow z = \frac{\frac{1}{2} \pm \sqrt{\frac{1}{4} - 1}}{2} = \frac{1}{4} \pm j\frac{\sqrt{3}}{4}$

$|z| = \sqrt{\frac{1}{16} + \frac{3}{16}} = \frac{1}{2} < 1 \Rightarrow$ as. stable

$v(t) = e(t) \Rightarrow$ SSP $\Rightarrow$ <u>$y(t)$ is SSP</u> $\Rightarrow$ Hyp is <u>correct</u>

$\mathbb{E}[y(t)^2] = \mathbb{E}[(\frac{1}{2}y(t-1) - \frac{1}{4}y(t-2) + e(t))^2]$

$= \frac{1}{4}\mathbb{E}[y(t-1)^2] + \frac{1}{16}\mathbb{E}[y(t-2)^2] + \mathbb{E}[e(t)^2] +$

↱ $f(e(t-2), e(t-3), \ldots)$

$- \frac{1}{4}\mathbb{E}[y(t-1)y(t-2)] + \mathbb{E}[y(t-1)e(t)] - \frac{1}{2}\mathbb{E}[y(t-2)e(t)]$

↳ $f(e(t-1), e(t-2), \ldots)$

$\gamma_y(0) = \frac{1}{4}\gamma_y(0) + \frac{1}{16}\gamma_y(0) + 1 - \frac{1}{4}\gamma_y(1) + 0 - 0 =$

$= \frac{5}{16}\gamma_y(0) - \frac{1}{4}\gamma_y(1) + 1$

$$\mathbb{E}[y(t)y(t-1)] = \mathbb{E}\left[\left(\tfrac{1}{2}y(t-1) - \tfrac{1}{4}y(t-2) + e(t)\right)y(t-1)\right] =$$
$$= \tfrac{1}{2}\mathbb{E}[y(t-1)^2] - \tfrac{1}{4}\mathbb{E}[y(t-1)y(t-2)] + \cancel{\mathbb{E}[y(t-1)e(t)]}\,0$$
$$= \tfrac{1}{2}\gamma_y(0) - \tfrac{1}{2}\gamma_y(1)$$

$$\downarrow$$

$$\gamma_y(1) = \tfrac{1}{2}\gamma_y(0) - \tfrac{1}{2}\gamma_y(1)$$

$$\Rightarrow \begin{cases} \tfrac{11}{16}\gamma_y(0) + \tfrac{1}{4}\gamma_y(1) = 1 \\ -\tfrac{1}{2}\gamma_y(0) + \tfrac{5}{4}\gamma_y(1) = 0 \end{cases} \longrightarrow \begin{array}{l} \gamma_y(0) = \tfrac{80}{63} \\ \gamma_y(1) = \tfrac{32}{63} \end{array}$$

$$\gamma_y(2) = \mathbb{E}[y(t)y(t-2)] = \mathbb{E}\left[\left(\tfrac{1}{2}y(t-1) - \tfrac{1}{4}y(t-2) + e(t)\right)y(t-2)\right]$$
$$= \tfrac{1}{2}\mathbb{E}[y(t-1)y(t-2)] - \tfrac{1}{4}\mathbb{E}[y(t-2)^2] + \cancel{\mathbb{E}[y(t-2)e(t)]}\,0 =$$
$$= \tfrac{1}{2}\gamma_y(1) - \tfrac{1}{4}\gamma_y(0) = -\tfrac{4}{63}$$

$$\gamma_y(\tau) = \tfrac{1}{2}\gamma_y(\tau-1) - \tfrac{1}{4}\gamma_y(\tau-2)$$

es 3

MA(∞) representation of es2's $y(t)$

1. Long division:

$$\begin{array}{rl|l}
1 & - & 1 - \tfrac{1}{2}z^{-1} + \tfrac{1}{4}z^{-2} \\
1 - \tfrac{1}{2}z^{-1} + \tfrac{1}{4}z^{-2} & = & 1 + \tfrac{1}{2}z^{-1} + 0z^{-2} - \tfrac{1}{8}z^{-3} \\
\hline
/ + \tfrac{1}{2}z^{-1} - \tfrac{1}{4}z^{-2} & - & \\
\tfrac{1}{2}z^{-2} + \tfrac{1}{4}z^{-2} + \tfrac{1}{8}z^{-3} & = & \\
\hline
/ \quad 0z^{-2} - \tfrac{1}{8}z^{-3} & - & \\
0z^{-2} + 0z^{-3} + 0z^{-4} & = & \\
\hline
/ \quad \tfrac{1}{8}z^{-3} \quad 0z^{-4} & & \\
\vdots & &
\end{array}$$

$$y(t) = 1\cdot e(t) + \tfrac{1}{2}e(t-1) + 0e(t-2) - \tfrac{1}{8}e(t-3) + \cdots$$

2. Substitution:

$$y(t) = \tfrac{1}{2}\underbrace{y(t-1)}_{\hookrightarrow\ \frac{1}{2}y(t-2) - \frac{1}{4}y(t-3) + e(t-1)} - \tfrac{1}{4}y(t-2) + e(t) =$$
$$= \tfrac{1}{2}\left(\tfrac{1}{2}y(t-2) - \tfrac{1}{4}y(t-3) + e(t-1)\right) - \tfrac{1}{4}y(t-2) + e(t) =$$
$$= \cancel{\tfrac{1}{4}y(t-2)} - \tfrac{1}{8}y(t-3) + \tfrac{1}{2}e(t-1) - \cancel{\tfrac{1}{4}y(t-2)} + e(t) =$$
$$= -\tfrac{1}{8}y(t-3) + e(t) + \tfrac{1}{2}e(t-1) =$$

$$= -\frac{1}{8}\left(\frac{1}{2}y(t-4) - \frac{1}{4}y(t-5) + e(t-3)\right) + e(t) + \frac{1}{2}e(t-1) =$$

$$= -\frac{1}{16}y(t-4) + \frac{1}{32}y(t-5) + \underbrace{e(t) + \frac{1}{2}e(t-1) - \frac{1}{18}e(t-3)}_{\text{partial representation}} =$$

$$\Rightarrow y(t-\underset{3}{\underline{1}}) = e(t-1) + \frac{1}{2}e(t-2) - \frac{1}{18}e(t-4)$$

02/03/23

es 1

$$y(t) = \frac{1}{3}y(t-1) + \underbrace{e(t)+2}_{v(t)} \qquad e(t) \sim WN(1,1)$$

$$y \sim AR(1)$$

$$y(t) = \frac{1}{3}z^{-1}y(t) + e(t) + 2 \quad \rightarrow \quad y(t) = \frac{1}{1-\frac{1}{3}z^{-1}}\underbrace{(e(t)+2)}_{\text{stationary}}$$

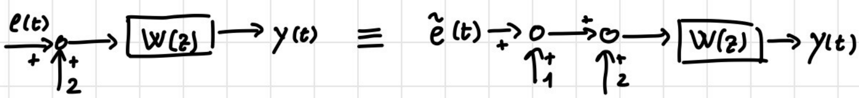
sing. of $W(z)$: $\frac{z}{z} - \frac{1}{3} = 0$ $\nearrow z_0 = 0$, $\searrow z_p = \frac{1}{3}$ $\Rightarrow$ $y(t)$ is stationary

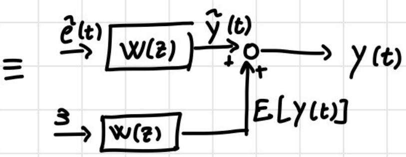
$$\mathbb{E}[y(t)] = \frac{1}{3}\mathbb{E}[y(t-1)] + 1 + 2 =$$
$$= \frac{1}{3}m + 3$$
$$\Rightarrow m = \frac{9}{2}$$

$$\tilde{y}(t) + \cancel{\frac{9}{2}} = \frac{1}{3}\left(\tilde{y}(t-1) + \frac{9}{2}\right) + \tilde{e}(t) + 1 + 2 \rightarrow \tilde{y}(t) = \frac{1}{3}\tilde{y}(t-1) + \tilde{e}(t)$$
$$= \frac{1}{3}\tilde{y}(t-1) + \cancel{\frac{3}{2}} + \tilde{e}(t) + \cancel{\frac{6}{2}}$$

$$\gamma_y(0) = \mathbb{E}[\tilde{y}(t)^2] = \frac{1}{9}\mathbb{E}[\tilde{y}(t-1)^2] + \mathbb{E}[e(t)^2] + \cancel{\frac{2}{3}[\tilde{y}(t-1)\tilde{e}(t)]}$$
$$= \frac{1}{9}\gamma_y(0) + \lambda^2 = \frac{1}{9}\gamma_y(0) + 1 = \frac{1}{1-\frac{1}{9}} = \frac{9}{8}$$

$$\gamma_y(1) = \mathbb{E}[\tilde{y}(t)\tilde{y}(t-1)] =$$
$$= \mathbb{E}\left[\left(\frac{1}{3}\tilde{y}(t-1) + \tilde{e}(t)\right)\tilde{y}(t-1)\right]$$
$$= \frac{1}{3}\gamma_y(0) + \cancel{\mathbb{E}[\tilde{e}(t)\tilde{y}(t-1)]} = \frac{1}{3}\cdot\frac{9}{8} = \frac{3}{8}$$





Final value theorem: If $W(z)$ is stable:

$$y(t) = \lim_{z\to 1} \frac{z-1}{z} W(z) \cdot \frac{z}{z-1} \cdot 3 = W(1)\cdot 3 \qquad ???$$

$$\Rightarrow \mathbb{E}[y(t)] = W(1)\,\mathbb{E}[e(t)]$$

es 2

$$y(t) = \frac{1}{2} y(t-1) + \eta(t) - \eta(t-1) \qquad \eta(t) \sim WN(1,9)$$
$$y(t) \sim ARMA(1,1)$$

$$y(t) = \frac{1}{2} z^{-1} y(t) + \eta(t) - z^{-1}\eta(t) = \frac{1-z^{-1}}{1-\frac{1}{2}z^{-1}}\eta(t)$$

singularities: $z-1=0 \to z_0 = 1$, $z - \frac{1}{2} = 0 \to z_p = \frac{1}{2}$ $\Rightarrow y(t)$ is stat.

$$\mathbb{E}[y(t)] = W(1)\,\mathbb{E}[\eta(t)] = \frac{1-1}{1-\frac{1}{2}} \cdot 1 = 0$$

$$e(t) = \eta(t) - \mathbb{E}[\eta(t)] \to y(t) = \frac{1}{2} y(t-1) + e(t) \cancel{+1} - e(t-1) \cancel{-1} =$$
$$= \frac{1}{2} y(t-1) + e(t) - e(t-1)$$

$$\gamma_y(0) = \mathbb{E}[y(t)^2] = \frac{1}{4}\mathbb{E}[y(t-1)^2] + \mathbb{E}[e(t)^2] + \mathbb{E}[e(t-1)^2] +$$
$$+ \cancel{\mathbb{E}[y(t-1)e(t)]} - \mathbb{E}[y(t-1)e(t-1)] \cancel{-2\mathbb{E}[e(t)e(t-1)]} =$$
$$= \frac{1}{4}\gamma_y(0) + \lambda^2 + \lambda^2 - \mathbb{E}[y(t-1)e(t-1)] =$$

$$\mathbb{E}[y(t-1)e(t-1)] = \mathbb{E}[(\tfrac{1}{2} y(t-2) + e(t-1) - e(t-2))\,e(t-1)]$$
$$= \cancel{\tfrac{1}{2}\mathbb{E}[y(t-2)e(t-1)]} + \lambda^2 - \cancel{\mathbb{E}[e(t-2)e(t-1)]} = \lambda^2$$

$$= \frac{1}{4}\gamma_y(0) + 9 = 9 \cdot \frac{4}{3} = 12$$

$$\gamma_y(1) = \mathbb{E}[y(t)y(t-1)] = \mathbb{E}[(\tfrac{1}{2} y(t-1) + e(t) - e(t-1))\,y(t-1)] =$$
$$= \frac{1}{2}\gamma_y(0) + \cancel{\mathbb{E}[e(t)y(t-1)]} - \mathbb{E}[e(t-1)y(t-1)] =$$
$$= \frac{1}{2}\gamma_y(0) - \lambda^2 = 6 - 9 = -3$$

$$\gamma_y(2) = \mathbb{E}[y(t)y(t-2)] = \mathbb{E}[(\tfrac{1}{2} y(t-1) + e(t) + e(t-1))\,y(t-2)] =$$
$$= \frac{1}{2}\gamma_y(1) + \cancel{\mathbb{E}[\dots]} + \cancel{\mathbb{E}[\dots]} = -\frac{3}{2}$$

$$\vdots$$

$$\gamma_y(\tau) = \frac{1}{2}\gamma_y(\tau-1) \qquad \tau \geq 2$$

es 3

$$y(t) = 0.5\, e(t-1) + 0.5\, e(t-2) + e(t)$$

$e(t) \sim WN(2,1)$

$y \sim MA(2)$

$$y(t) = (0.5\, z^{-1} + 0.5\, z^{-2} + 1)\, e(t)$$
$$= \frac{0.5 + 0.5 z + z^2}{z^2}\, e(t)$$

$$\tilde{e}(t) = e(t) - 2$$

$$\rightarrow y(t) = 0.5(\tilde{e}(t-1)+2) + 0.5(\tilde{e}(t-2)+2) + \tilde{e}(t) + 2 =$$
$$= 0.5\tilde{e}(t-1) + 1 + 0.5\tilde{e}(t-2) + 1 + \tilde{e}(t) + 2 =$$
$$= 0.5\tilde{e}(t-1) + 0.5\tilde{e}(t-2) + \tilde{e}(t) + 4$$

$$\mathbb{E}[y(t)] = \frac{1/2 + 1/2 + 1}{1} \cdot 2 = 4$$

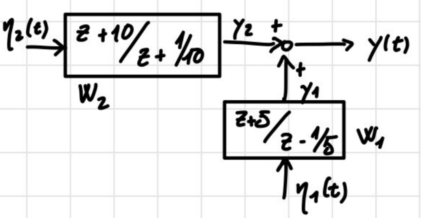
$$\tilde{y}(t) = y(t) - 4$$

$$\rightarrow \tilde{y}(t) = 0.5\, e(t-1) + 0.5\tilde{e}(t-2) + \tilde{e}(t)$$

$$\gamma_y(\tau) = \begin{cases} (c_0^2 + c_1^2 + c_2^2)\lambda^2 = 1.5 & \tau = 0 \\ (c_0 c_1 + c_1 c_2)\lambda^2 = 0.75 & \tau = \pm 1 \\ c_0 c_2 \lambda^2 = 0.5 & \tau = \pm 2 \\ 0 & |\tau| > 2 \end{cases}$$

06/03/23

es 1



$\left.\begin{matrix} \eta_1 \sim W(1,1) \\ \eta_2 \sim W(0,1) \end{matrix}\right\} \rightarrow$ uncorrelated

$\mathbb{E}[(\eta_1(t_1) - \mu_{\eta_1})(\eta_2(t_2) - \mu_{\eta_2})] = 0 \quad \forall t_1, t_2$

$y(t)$ is SSP?

$$y(t) = y_1(t) + y_2(t)$$

Since η_1 and η_2 are uncorrelated, then y_1 and y_2 are also uncorrelated. Since the poles of the two tranfers are inside the unit circle, y_1 and y_2 are SSPs. Thus $y(t)$ is also an SSP.

$$\mathbb{E}[y(t)] = \mathbb{E}[y_1(t)] + \mathbb{E}[y_2(t)] = W_1(1)\,\mathbb{E}[\eta_1(t)] + W_2(1)\,\mathbb{E}[\eta_2(t)] =$$
$$= \dots = 15/2$$

y_1 and y_2 are uncorrelated

$$\gamma_y(\tau) = \gamma_{y_1}(\tau) + \gamma_{y_2}(\tau)$$

1

$$y_1(t) = \frac{1+5z^{-1}}{1-1/5\,z^{-1}}\,\eta_1(t) = 1/5\,y_1(t-1) + \eta_1(t) + 5\,\eta_1(t-1)$$

$$\tilde{y}_1(t) = y_1(t) - 15/2 \qquad \tilde{\eta}_1(t) = \eta_1(t) - 1$$

$$\tilde{y}_1(t) + 15/2 = 1/5\left(\tilde{y}_1(t-1) + 15/2\right) + \tilde{\eta}_1(t) + 1 + 5\,\tilde{\eta}_1(t) + 5$$
$$\Rightarrow \tilde{y}_1(t) = 1/5\,\tilde{y}_1(t-1) + \tilde{\eta}_1(t) + 5\,\tilde{\eta}_1(t-1)$$

$$\gamma_{y_1}(0) = \mathbb{E}[\tilde{y}_1(t)^2] = \mathbb{E}[(1/5\,\tilde{y}_1(t-1) + \tilde{\eta}_1(t) + 5\tilde{\eta}_1(t-1))^2]$$
$$= 1/25\,\mathbb{E}[\tilde{y}(t-1)^2] + \mathbb{E}[\tilde{\eta}_1(t)^2] + 25\,\mathbb{E}[\tilde{\eta}_1(t-1)^2] +$$
$$+ \cancel{2/5\,\mathbb{E}[\tilde{y}_1(t-1)\,\eta_1(t)]} + 2\,\mathbb{E}[\tilde{y}_1(t-1)\,\tilde{\eta}(t-1)] + \cancel{10\,\mathbb{E}[\tilde{\eta}_1(t)\,\tilde{\eta}_1(t-1)]} =$$
$$= 1/25\,\gamma_{y_1}(0) + 1 + 25 + 2\,\mathbb{E}[\tilde{y}_1(t-1)\,\tilde{\eta}_1(t-1)]$$

$$\mathbb{E}[\tilde{y}_1(t-1)\,\tilde{\eta}_1(t-1)] = \cancel{1/5\,\mathbb{E}[\tilde{y}_1(t-2)\,\tilde{\eta}_1(t-1)]} + \mathbb{E}[\tilde{\eta}_1(t-1)^2] \cancel{+5\,\mathbb{E}[\tilde{\eta}_1(t-2)\,\tilde{\eta}_1(t-1)]}$$
$$= 1$$

$$= 1/25\,\gamma_{y_1}(0) + 1 + 25 + 2\cdot 1 = \dots = 175/6$$

$$\gamma_{y_1}(1) = \mathbb{E}[\tilde{y}_1(t)\,\tilde{y}_1(t-1)] = \dots = 65/6$$
$$\gamma_{y_1}(2) = \mathbb{E}[\tilde{y}_1(t)\,\tilde{y}_1(t-2)] = \dots = 13/6$$
$$\vdots$$
$$\gamma_{y_1}(\tau) = 1/5\,\gamma_{y_1}(\tau-1) \quad |\tau| \geq 2$$

2

$$y_2(t) = \frac{1+10z^{-1}}{1+1/10\,z^{-1}}\,\eta_2(t) = -1/10\,y_2(t-1) + \eta_2(t) + 10\,\eta_2(t-1)$$

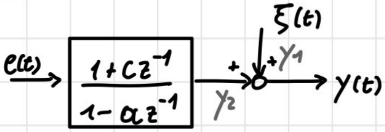
$$\gamma_{y_2}(0) = \mathbb{E}[y_2(t)^2] = \dots = 100$$
$$\vdots$$
$$\gamma_{y_2}(\tau) = 0 \quad \forall \tau \geq 1$$

In general:

$$e(t) \rightarrow \boxed{\frac{z+a}{z+1/a}} \rightarrow y(t) \qquad e(t) \sim WN(0, \lambda^2)$$
$$\Rightarrow y(t) \sim WN(0, a^2\lambda^2)$$

$\Rightarrow$ ALL-PASS FILTER

es 2



$\left.\begin{array}{l} e(t) \sim WN(0,1) \\ \xi(t) \sim WN(0,1) \end{array}\right\} \rightarrow$ uncorrelated

1. Choose c and a such that y(t) is stationary: $|a| < 1$
2. Given $\gamma_y(0) = 6$, $\gamma_y(1) = -2$, $\gamma_y(\tau) = 0$ $|\tau| > 1$, which c and a were chosen

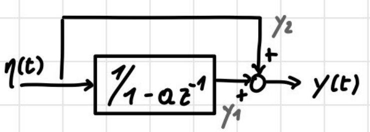
Due to the values, we can infer a MA(1) $\Rightarrow a = 0$

$$y(t) = e(t) + ce(t-1) + \xi(t)$$

$$\gamma_y(0) = \mathbb{E}[(e(t) + ce(t-1) + \xi(t))^2] =$$
$$= \mathbb{E}[e(t)^2] + c^2\mathbb{E}[e(t-1)^2] + \mathbb{E}[\xi(t)^2] +$$
$$+ \cancel{2c\,\mathbb{E}[e(t)e(t-1)]} + \cancel{2\mathbb{E}[e(t)\xi(t)]} + \cancel{2c\,\mathbb{E}[e(t-1)\xi(t)]} =$$
$$= 1 + c^2 + 1 = 6$$
$$\Rightarrow c = \pm 2$$

$$\gamma_y(1) = \mathbb{E}[y(t)y(t-1)] = \mathbb{E}[(e(t) + ce(t-1) + \xi(t))(e(t-1) + ce(t-2) + \xi(t-1))] =$$
$$= c\,\mathbb{E}[e(t-1)^2] = c = -2$$
$$\Rightarrow \underline{c = -2}$$

es 3



$|a| < 1$

$\eta(t) \sim WN(0, \lambda^2)$

$$\mathbb{E}[y(t)] = \mathbb{E}[y_1(t)] + \cancel{\mathbb{E}[y_2(t)]} = \mathbb{E}[y_1(t)] = W(1) \cdot 0 = 0$$

$$\gamma_y(0) = \mathbb{E}[y(t)^2] = \mathbb{E}[(y_1(t) + y_2(t))^2] = \mathbb{E}[y_1(t)^2] + \lambda^2 + 2\mathbb{E}[y_1(t)y_2(t)]$$

$y_1(t) = a\,y_1(t-1) + \eta(t)$

$y_2(t) = \eta(t)$

$$= 1/1-a^2\,\lambda^2 + \lambda^2 + 2(\cancel{\mathbb{E}[a\,y(t-1)\,\eta(t)]} + \lambda^2) = \frac{4-3a^2}{1-a^2}\lambda^2$$

$$\gamma_y(1) = \mathbb{E}[(y_1(t)+y_2(t))(y_1(t-1)+y_2(t-1))] =$$
$$= \mathbb{E}[y_1(t)y_1(t-1)] + \mathbb{E}[y_1(t)y_2(t-1)] + \mathbb{E}[y_2(t)y_1(t-1)] + \cancel{\mathbb{E}[y_2(t)y_2(t-1)]} =$$
$$= a \cdot \frac{1}{1-a^2}\lambda^2 + \mathbb{E}[(a y_1(t-1)+\eta(t))\eta(t-1)] + \cancel{\mathbb{E}[\eta(t)(a y_1(t-2)+\eta(t-1))]} =$$
$$= \frac{a}{1-a^2}\lambda^2 + a\,\mathbb{E}[y_1(t-1)\eta(t-1)]$$
$$= \frac{a}{1-a^2}\lambda^2 + a\left(\cancel{a\,\mathbb{E}[y_1(t-2)\eta(t-1)]} + \mathbb{E}[\eta(t-1)^2]\right) =$$
$$= \frac{a}{1-a^2}\lambda^2 + a\lambda^2 = \frac{a\lambda^2(2-a^2)}{1-a^2}$$

$$\gamma_y(2) = \dots = \frac{a^2\lambda^2}{1-a^2} + a^2\lambda^2$$
$$\vdots$$
$$\gamma_y(\tau) = a^\tau \frac{2-a^2}{1-a^2}\lambda^2$$

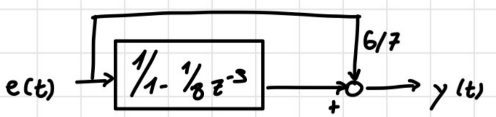
Alternative method:

$$y(t) = \frac{1}{1-az^{-1}}\eta(t) + \eta(t) = \frac{1+1-az^{-1}}{1-az^{-1}}\eta(t) = \frac{2-az^{-1}}{1-az^{-1}}\eta(t) = \frac{1-\frac{a}{2}z^{-1}}{1-az^{-1}}2\eta(t)$$
$$= \frac{1-\frac{a}{2}z^{-1}}{1-az^{-1}}\eta_1(t) \qquad \eta_1(t) \sim WN(0, 2^2\lambda^2)$$

$\Rightarrow y(t) = a y(t-1) + \eta_1(t) - \frac{a}{2}\eta_1(t-1)$ (ARMA(1,1))

$\rightarrow$ compute γ from here $\rightarrow$ it will lead to a simpler computation

Homework



$e(t) \sim WN(1, 63/64)$

$$\left[\begin{array}{l} \mathbb{E}[y(t)] = 2 \\ \gamma_y(0) = 1 \\ \gamma_y(1) = \gamma_y(2) = 0 \\ \gamma_y(\tau) = \frac{1}{8}\gamma(\tau-3) \quad |\tau| \geq 3 \end{array}\right]$$

15/03/23

esercizio 1

$$y(t) = \frac{1}{2}\, y(t-1) + e(t) - e(t-1) \qquad e(t) \sim WN(0,9)$$

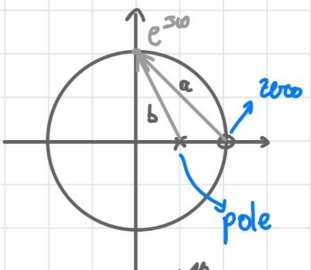
$$y(t) = \frac{1-z^{-1}}{1-\frac{1}{2}z^{-1}}\, e(t) = \frac{z-1}{z-\frac{1}{2}}\, e(t) \implies \Gamma_y(\omega) = \frac{|e^{j\omega}-1|^2}{|e^{j\omega}-\frac{1}{2}|^2}\cdot 9 = \frac{(e^{j\omega}-1)(e^{-j\omega}-1)\,9}{(e^{j\omega}-\frac{1}{2})(e^{-j\omega}-\frac{1}{2})}$$

$$= \frac{(1-e^{j\omega}-e^{-j\omega}+1)\,9}{(1-\frac{1}{2}e^{j\omega}-\frac{1}{2}e^{-j\omega}+\frac{1}{4})}$$

$$= \frac{(2-(e^{j\omega}+e^{-j\omega}))\,9}{\left(\frac{5}{4}-\left(\frac{e^{j\omega}}{2}+\frac{e^{j\omega}}{2}\right)\right)} =$$

$$= \frac{18-18\cos\omega}{\frac{5}{4}-\cos\omega}$$

Graphical method

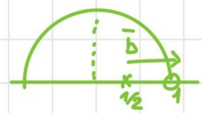


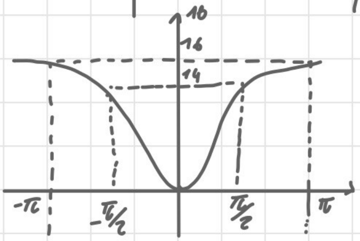
$$\Gamma_y(\omega) = \frac{|a|^2}{|b|^2}\cdot 9$$

$$\Gamma_y(0) = \frac{|0|^2}{(\dots)}\cdot 9 = 0$$

$$\Gamma_y\left(\frac{\pi}{2}\right) = \frac{1+1}{\frac{1}{4}+1}\cdot 9 = \frac{72}{5} \sim 14$$

$$\Gamma_y(\pi) = \frac{2^2}{(1+\frac{1}{2})^2}\cdot 9 = 16$$





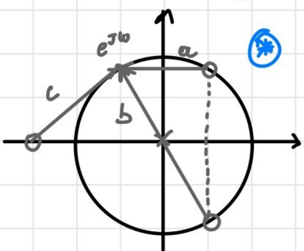
$\implies$ hi-pass filter

exercise 2

$$y(t) = (1-z^{-1}+z^{-2})(1+\tfrac{3}{2}z^{-1})\,e(t) \qquad e(t) \sim WN(0,1)$$

$$= \frac{(z^2-z+1)(z+\frac{3}{2})}{z^3}\,e(t) \longrightarrow$$ poles: $z=0$

zeroes: $z_1 = -\frac{3}{2}$ $\quad z_{2,3} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}$ $\qquad |z_{2,3}| = 1$

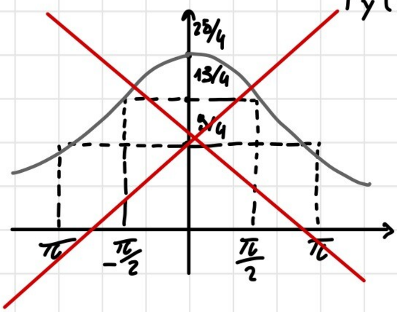


$$\Gamma_y(\omega) = \frac{|a|^2|b|^2|c|^2}{1}\cdot 1$$

$$\Gamma_y(0) = \underbrace{\cancel{(\tfrac{1}{4}+\tfrac{3}{4})}}_{1}\underbrace{\cancel{(\tfrac{1}{4}+\tfrac{3}{4})}}_{1}(2+\tfrac{1}{2})^2 = \frac{25}{4}$$

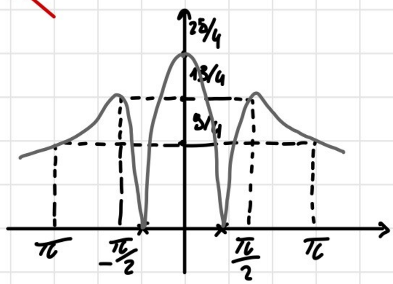
$$\Gamma_y\left(\tfrac{\pi}{2}\right) = \left(\tfrac{1}{4} + \left(1 - \tfrac{\sqrt{3}}{2}\right)^2\right)\left(\tfrac{1}{4} + \left(1 + \tfrac{\sqrt{3}}{2}\right)^2\right)\left(1 + \left(1 + \tfrac{1}{2}\right)^2\right)$$
$$= (2-\sqrt{3})(2+\sqrt{3}) \cdot \tfrac{13}{4} = (4-3)\,\tfrac{13}{4} = \tfrac{13}{4}$$

$$\Gamma_y(\pi) = \dots = \tfrac{9}{4}$$



WRONG: The spectrum is zero in $\pm \frac{\pi}{3}$

Notice the zeroes on the unit circle in (*)



16/03/23

exercise 1

$$y(t) = \tfrac{1}{2}\, y(t-2) + \eta(t) + 4\,\eta(t-1) \qquad \eta(t) \sim WN(0,1)$$

$$y(t) = \frac{1+4z^{-1}}{1-\frac{1}{2}z^{-2}}\,\eta(t) = \frac{z(z+4)}{z^2-\frac{1}{2}}\,\eta(t) \qquad \hat{y}(t+1|t) = ??$$

STEP 1: Check if $W(z)$ is canonical.

1. $N(z)$ and $D(z)$ must be monic ✓
2. $N(z)$ and $D(z)$ coprime ✓
3. $N(z)$ and $D(z)$ of the same degree ✓
4. $N(z)$ and $D(z)$ singularities must be inside the unit circle ✗

zeroes: $z=0$, $\underline{z=-4}$ → we need to remove the zero

poles: $z_{1,2} = \frac{1}{\sqrt{2}}$ $\quad |z_{1,2}| < 1$

$$y(t) = \frac{1+4z^{-1}}{1-\frac{1}{2}z^{-2}}\,\eta(t) = \frac{1+\frac{1}{4}z^{-1}}{1+\frac{1}{4}z^{-1}} \cdot \frac{1+4z^{-1}}{1-\frac{1}{2}z^{-2}}\,\eta(t) = \frac{1+\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}} \cdot \underbrace{\frac{1+4z^{-1}}{1+\frac{1}{4}z^{-1}}}_{\text{ALL-PASS FILTER}}\,\eta(t)$$

$$= \frac{1+\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-2}}\, e(t) \qquad e(t) \sim WN(0,16)$$

STEP 2: Computation of predictor

1. Long division:

$$\begin{array}{l|l} 1 + \frac{1}{4} z^{-1} & 1 + 0 z^{-1} - \frac{1}{2} z^{-2} \\ \hline -1 - 0 z^{-1} + \frac{1}{2} z^{-2} & 1 \\ \hline / \ \frac{1}{4} z^{-1} + \frac{1}{2} z^{-2} & \end{array}$$

$$\Rightarrow y(t) = 1 \cdot e(t) + \frac{\frac{1}{4} z^{-1} + \frac{1}{2} z^{-2}}{1 - \frac{1}{2} z^{-2}} e(t)$$

$$= \underbrace{e(t)}_{\substack{\text{not predictable} \\ \text{b.c. WN with } \mu = 0}} + \frac{\frac{1}{4} + \frac{1}{2} z^{-1}}{1 - \frac{1}{2} z^{-2}} e(t-1)$$

$$\Rightarrow \hat{y}(t|t-1) = \frac{\frac{1}{4} + \frac{1}{2} z^{-1}}{1 - \frac{1}{2} z^{-2}} e(t-1) \rightarrow \text{predictor from noise}$$

$$= \frac{\frac{1}{4} z^{-1} + \frac{1}{2} z^{-2}}{\cancel{1 - \frac{1}{2} z^{-2}}} \underbrace{\frac{\cancel{1 - \frac{1}{2} z^{-2}}}{1 + \frac{1}{4} z^{-1}}}_{W(z)^{-1}} y(t) = \frac{\frac{1}{4} + \frac{1}{2} z^{-1}}{1 + \frac{1}{4} z^{-1}} y(t-1)$$

$$\hat{y}(t+1|t) = \frac{\frac{1}{4} + \frac{1}{2} z^{-1}}{1 + \frac{1}{4} z^{-1}} y(t) \rightarrow \hat{y}(t+1|t) = -\frac{1}{4} \hat{y}(t|t-1) + \frac{1}{4} y(t) + \frac{1}{2} y(t-1)$$

Simplified formula: $y(t) = \frac{C(z)}{A(z)} e(t) \rightarrow \hat{y}(t|t-1) = \frac{C(z) - A(z)}{A(z)} e(t)$

$$= \frac{C(z) - A(z)}{C(z)} y(t)$$

STEP 3: Check the variance of the predictor

$$\varepsilon(t|t-1) = y(t) - \hat{y}(t|t-1) = E(z) e(t) + \cancel{\frac{F(z)}{A(z)}} e(t) - \cancel{\frac{F(z)}{A(z)}} e(t) = E(z) e(t)$$

$\mathbb{E}[\varepsilon(t|t-1)] = \mathbb{E}[e(t)^2] = 16$ → the variance is that of the noise, meaning we have the best possible predictor

→ we also have variance near that of the process itself, which is pretty bad

$\hat{y}(t+2|t) = ??$

$$\begin{array}{l|l} 1 + \frac{1}{4} z^{-1} & 1 + 0 z^{-1} - \frac{1}{2} z^{-2} \\ \hline -1 - 0 z^{-1} + \frac{1}{2} z^{-2} & 1 + \frac{1}{4} z^{-1} \\ \hline / \ \frac{1}{4} z^{-1} + \frac{1}{2} z^{-2} & \\ -\frac{1}{4} z^{-1} - 0 z^{-2} + \frac{1}{8} z^{-3} & \\ \hline / \ \frac{1}{2} z^{-2} + \frac{1}{8} z^{-3} & \end{array}$$

$$\Rightarrow \hat{y}(t|t-2) = \frac{\frac{1}{2} z^{-2} + \frac{1}{8} z^{-3}}{1 - \frac{1}{2} z^{-2}} e(t)$$

$$= \frac{\frac{1}{2} z^{-2} + \frac{1}{8} z^{-3}}{\cancel{1 - \frac{1}{2} z^{-2}}} \cdot \frac{\cancel{1 - \frac{1}{2} z^{-2}}}{1 + \frac{1}{4} z^{-1}} y(t) = \frac{\frac{1}{2} + \frac{1}{8} z^{-3}}{1 + \frac{1}{4} z^{-1}} y(t-2)$$

$$\varepsilon(t|t-2) = (1 + \tfrac{1}{4} z^{-1}) e(t)$$

$$\mathbb{E}[\varepsilon(t|t-2)^2] = \mathbb{E}[(e(t) + \tfrac{1}{4} e(t-1))^2] = \dots = 17$$

→ if the horizon increases, we approach the equivalent MA with variance equal of that of the process.

$$\hat{y}(t|t-2) = \frac{\frac{1}{2} + \frac{1}{8} z^{-1}}{1 + \frac{1}{4} z^{-1}} y(t-2) = \frac{1}{2} \frac{\cancel{1 + \frac{1}{4} z^{-1}}}{\cancel{1 + \frac{1}{4} z^{-1}}} y(t-2) = \frac{1}{2} y(t-2)$$

20/03/23

exercise 1

$y(t) = e(t) + 5e(t-1)$ $\qquad e(t) \sim WN(1,1)$

$\qquad \mathbb{E}[y(t)] = 6$

↳ $\tilde{y}(t) = \tilde{e}(t) + 5e(t-1)$ $\qquad \tilde{e}(t) = e(t) - 1$

$$\tilde{y}(t) = (1 + 5z^{-1})\tilde{e}(t) = \frac{1 + \frac{1}{5} z^{-1}}{1} \cdot \underbrace{\frac{1 + 5z^{-1}}{1 + \frac{1}{5} z^{-1}} \tilde{e}(t)}_{WN(0,25)} = (1 + \tfrac{1}{5} z^{-1}) \eta(t)$$

$$\hat{y}(t|t-1) = \,? \xrightarrow{\text{DEBIAS}} \hat{\tilde{y}}(t|t-1) = \tfrac{1}{5} \eta(t-1) \quad \text{(from noise)}$$

$$= \frac{\frac{1}{5} z^{-1}}{1 + \frac{1}{5} z^{-1}} \tilde{y}(t)$$

$$\rightarrow \hat{\tilde{y}}(t|t-1) = -\tfrac{1}{5} \hat{\tilde{y}}(t-1|t-2) + \tfrac{1}{5} \tilde{y}(t-1)$$

$$\hat{y}(t|t-1) - 6 = -\tfrac{1}{5}(\hat{y}(t-1|t-2) - 6) + \tfrac{1}{5}(y(t-1) - 6)$$

$$\rightarrow \hat{y}(t|t-1) = -\tfrac{1}{5} \hat{y}(t-1|t-2) + \tfrac{1}{5} y(t-1) + 6$$

exercise 2

$$\tfrac{1}{2} y(t) = -\tfrac{1}{3} y(t-1) - \tfrac{1}{18} y(t-2) + 3e(t-2) - 8e(t-3) - 3e(t-4) \qquad e(t) \sim WN(0,1)$$

$$y(t) = \frac{3z^{-2} - 8z^{3} - 3z^{-4}}{\frac{1}{2} + \frac{1}{3} z^{-1} + \frac{1}{18} z^{-2}} e(t)$$

$$\hookrightarrow y(t) = \frac{3z^{-2}}{\frac{1}{2}} \cdot \frac{1 - \frac{8}{3} z^{-1} - z^{-2}}{1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}} e(t) = \frac{1 - \frac{8}{3} z^{-1} - z^{-2}}{1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}} \eta(t)$$

$$\hookrightarrow \eta(t) = \dots = 6e(t-1) \sim WN(0,36)$$

zeroes: $z_1 = 3 \qquad z = -1/3$

poles: $z_{1,2} = 1/3$

$$\Rightarrow y(t) = \frac{(1-3z^{-1})\cancel{(1+1/3 z^{-1})}}{\cancel{(1+1/3z^{-1})}(1+1/3 z^{-1})}\eta(t)$$

$$= \frac{1-1/3 z^{-1}}{1+1/3 z^{-1}} \underbrace{\frac{1-3z^{-1}}{1-1/3 z^{-1}}\eta(t)}_{\xi(t) \sim WN(0,\, 18^2)}$$

$$= \frac{1-1/3 z^{-1}}{1+1/3 z^{-1}} \xi(t)$$

$$\hat{y}(t|t-1) = \frac{C(z)-A(z)}{C(z)} = \dots = \frac{-2/3 z^{-1}}{1-1/3 z^{-1}} y(t)$$

exercise 3

$$y(t) = 1/3\, y(t-1) + u(t-1) + 3e(t-1) + e(t-2) \rightarrow \frac{C(z)}{A(z)} e(t) + \frac{B(z)}{A(z)} u(t-1)$$

$$y(t) = \underbrace{\frac{3z^{-1} + z^{-2}}{1-1/3 z^{-1}}}_{\text{we consider the transfer from noise}} e(t) + \frac{1}{1-1/3 z^{-1}} u(t-1)$$

$$W(z) = \frac{3z^{-1}}{1} \cdot \frac{1+1/3 z^{-1}}{1-1/3 z^{-1}} \longrightarrow \quad \eta(t) = 3z^{-1} e(t) = 3e(t-1) \sim WN(0,9)$$

$$\Rightarrow y(t) = \frac{1+1/3 z^{-1}}{1-1/3 z^{-1}} \eta(t) + \frac{1}{1-1/3 z^{-1}} u(t-1)$$

$$\hat{y}(t|t-1) = \frac{F(z)}{C(z)} y(t) + \frac{B(z)E(z)}{C(z)} u(t-1)$$

$$= \frac{2/3 z^{-1}}{1+1/3 z^{-1}} y(t) + \frac{1}{1+1/3 z^{-1}} u(t-1)$$

$$\begin{array}{r|l} 1 + 1/3 z^{-1} & 1 - 1/3 z^{-1} \\ \hline -1 + 1/3 z^{-1} & \underbrace{1}_{E(z)} \\ \hline 1 + \underbrace{2/3 z^{-1}}_{F(z)} & \end{array}$$

exercise 4

$$y(t) = 3 + v(t) \qquad \hat{y}(t|t-k) = ? \quad \forall k$$

1. $v(t) \sim WN(0,1)$
2. $v(t) = e(t) + 1/2\, e(t-2)$

1. $\hat{y}(t|t-k) = \mathbb{E}[y(t)] = 3$
2. $y(t) = 3 + e(t) + 1/2\, e(t-2) \qquad \mathbb{E}[y(t)] = 3$

$$\rightarrow \tilde{y}(t) = e(t) + 1/2\, e(t-2)$$

$$\tilde{y} = \frac{1 + 1/2 z^{-2}}{1} e(t)$$

$$
\begin{array}{l|l}
1 + 0 + \frac{1}{2} z^{-2} & 1 \\ \cline{2-2}
-1 & 1 + 0 z^{-1} + \frac{1}{2} z^{-2} \\ \cline{1-1}
\text{F@K=1} \quad / \quad 0 + \frac{1}{2} z^{-2} & \\
\quad - 0 z^{-1} & \\ \cline{1-1}
\text{F@ K=2} \quad / \quad + \frac{1}{2} z^{-2} & \\
\quad - \frac{1}{2} z^{-2} & \\ \cline{1-1}
\text{F@ K}\geq 3 \quad / &
\end{array}
$$

$\underline{K=1}$

$$\hat{\tilde{y}}(t|t-1) = \frac{\frac{1}{2} z^{-2}}{1 + \frac{1}{2} z^{-2}} \tilde{y}(t)$$

$$= -\frac{1}{2} \hat{\tilde{y}}(t-1|t-2) + \frac{1}{2} \tilde{y}(t-2)$$

$$\hat{y}(t|t-1) = -\frac{1}{2} \hat{y}(t-1|t-2) + \frac{1}{2} y(t-2) + 3$$

$\underline{K=2}$

$$\hat{y}(t|t-2) = -\frac{1}{2} \hat{y}(t-2|t-4) + \frac{1}{2} y(t-2) + 3$$

$\underline{K>3}$ $\quad \hat{y}(t|t-K) = 3$

exercise 5

$$y(t) = \frac{1}{4} y(t-2) + \eta(t-1) + \frac{1}{3} \eta(t-3) \qquad \eta(t) \sim WN(0,1)$$

$\hat{y}(t|t-2) = ?$

$$y(t) = \frac{z^{-1} + \frac{1}{3} z^{-3}}{1 - \frac{1}{4} z^{-2}} \eta(t) = \frac{1 + \frac{1}{3} z^{-2}}{1 - \frac{1}{4} z^{-2}} z^{-1} \eta(t) = \frac{1 + \frac{1}{3} z^{-2}}{1 - \frac{1}{4} z^{-2}} e(t)$$

zeroes : $\quad z_{1,2} = \pm j \frac{\sqrt{3}}{3}$

poles : $\quad z_1 = 0 \qquad z_{2,3} = \pm \frac{1}{2}$

$\hookrightarrow$ for the factorized z^{-1}

$$
\begin{array}{l|l}
1 + 0 z^{-1} + \frac{1}{3} z^{-2} & 1 + 0 z^{-1} - \frac{1}{4} z^{-2} \\ \cline{2-2}
-1 - 0 \quad + \frac{1}{4} z^{-2} & \underline{1 + 0 z^{1}} \\ \cline{1-1}
/ \quad 0 \quad + \frac{7}{12} z^{-2} & \\
\quad 0 \quad + 0 & \\ \cline{1-1}
\quad / \quad + \underline{\frac{7}{12} z^{-2}} &
\end{array}
\quad \Longrightarrow \quad y(t) = \underbrace{e(t)}_{\text{unpredictable}} + \frac{7/12}{1 - \frac{1}{4} z^{-2}} e(t-2)
$$

$$\rightarrow \hat{y}(t) = \frac{\frac{7}{12} z^{-2}}{1 - \frac{1}{4} z^{-2}} e(t)$$

$$= \frac{\frac{7}{12} z^{-2}}{\cancel{1 - \frac{1}{4} z^{-2}}} \cdot \frac{\cancel{1 - \frac{1}{4} z^{-2}}}{1 + \frac{1}{3} z^{-2}} y(t)$$

$$= \frac{7/12}{1 + \frac{1}{3} z^{-2}} y(t-2)$$

21/03/23

exercise 1

$\mathcal{S}: \; y(t) = e(t) + \frac{1}{2} e(t-1) \qquad e(t) \sim WN(0,1) \qquad MA(1)$

$m: \; y(t) = a\, y(t-1) + c(t) \qquad c(t) \sim WN(0, \sigma^2) \qquad AR(1)$

① $\hat{m}: \quad \hat{y}(t|t-1) = a\, y(t-1)$

② $\varepsilon: \quad \varepsilon(t|t-1) = y(t) - a y(t-1) = (1 - a z^{-1})\, y(t)$

$$= (1 - a z^{-1})(1 + \tfrac{1}{2} z^{-1})\, e(t)$$

$$= e(t) + (\tfrac{1}{2} - a)\, e(t-1) - \tfrac{1}{2} a\, e(t-2)$$

③ $\mathbb{E}[\varepsilon^2]: \quad \mathbb{E}[\varepsilon(t|t-1)^2] = 1 + (\tfrac{1}{2} - a)^2 + \tfrac{1}{4} a^2$

$$= 1 + \tfrac{1}{4} + a^2 - a + \tfrac{1}{4} a^2 = \tfrac{5}{4} + \tfrac{5}{4} a^2 - a = \hat{J}(a$$

④ $\frac{d}{da} \bar{J}(a) = \frac{5}{2} a - 1 = 0 \quad \Rightarrow \quad a^* = \frac{2}{5}$

⑤ $\delta^{*2} = \bar{J}(a^*) = \dots = \frac{21}{20} \sim 1$ → variance ~1 indicates a good model

→ $\varepsilon(t|t-1) = e(t) + \frac{1}{10} e(t-1) + \frac{1}{5} e(t-2)$ → error that is more similar / equal to WN indicates a good model

exercise 2

$\mathcal{S}: \; y(t) = e(t) + \frac{1}{2} e(t-1) \qquad e(t) \sim WN(0,1) \qquad MA(1)$

$m: \; y(t) = \eta(t) + b\, \eta(t-1) \qquad \eta(t) \sim WN(0, \lambda^2) \qquad MA(1)$

$\Rightarrow \; b^* = \frac{1}{2} \quad \lambda^{*2} = 1$ → model is of the same class as target

exercise 3

$\mathcal{S}: \; y(t) = e(t) + \frac{1}{2} e(t-1) \qquad e(t) \sim WN(0,1) \qquad \theta^* = \begin{bmatrix} a^* \\ b^* \end{bmatrix} \quad \lambda^{2*}$

$m: \; y(t) = \left(\frac{1}{1 + a z^{-1} + b z^{-2}}\right) \eta(t)$

$$\hat{y}(t|t-1) = \frac{C_m(z) - A_m(z)}{C_m(z)}\, y(t)$$

$$\varepsilon(t|t-1) = \frac{A_m(z)}{C_m(z)}\, y(t) = \frac{1 + a z^{-1} + b z^{-1}}{1} \left(1 + \tfrac{1}{2} z^{-1}\right) e(t) = \dots$$

$$= \left(1 + (\tfrac{1}{2} + a) z^{-1} + (b + \tfrac{a}{2}) z^{-2} + \tfrac{b}{2} z^{-3}\right) e(t)$$

$$J(\theta) = 1 + (\tfrac{1}{2} + a)^2 + (b + \tfrac{a}{2})^2 + \tfrac{b^2}{4} = \tfrac{5}{4} a^2 + \tfrac{5}{4} b^2 + a + ab + \tfrac{5}{4}$$

$$\begin{cases} \frac{\partial}{\partial a} J(\theta) = 2 \cdot \frac{5}{4} a + 1 + b = 0 \\ \frac{\partial}{\partial b} J(\theta) = 2 \cdot \frac{5}{4} b + a = 0 \end{cases} \rightarrow \begin{cases} a^* = -\frac{10}{21} \\ b^* = \frac{4}{21} \end{cases}$$

Check stationarity of the model: $z^2 - \frac{10}{21} z + \frac{4}{21} = 0 \quad |z_{1,2}| = 0.4365$
$\Longrightarrow$ OK

$$\varepsilon(t) = e(t) - \left(\frac{1}{2} - \frac{10}{21}\right) e(t-1) + \left(\frac{4}{21} - \frac{5}{21}\right) e(t-1) + \frac{2}{21} e(t-3) \qquad \varepsilon(t|t-1)\big|_{\theta^*}$$
$$\lambda^{*2} = \dots = 1.011 = \mathbb{E}[\varepsilon(t)^2]$$

exercise 4

$S: \ y(t) = 3e(t) + 9e(t-1) \qquad e(t) \sim WN(0,1) \qquad \theta^* = [b^*] \quad \lambda^{2*}$
$m: \ y(t) = \eta(t) + b\eta(t-1) \qquad \eta(t) \sim WN(0,\lambda^2)$

System is not written in canonical form, let's rewrite it in canonical

$$S: \ y(t) = 3(e(t) + 3e(t-1)) = 3(1 + 3z^{-1})e(t) = \left(3\frac{1+3z^{-1}}{1+\frac{1}{3}z^{-1}}\left(1+\frac{1}{3}z^{-1}\right)\right)e(t)$$
$$= \left(1 + \frac{1}{3}z^{-1}\right)\xi(t)$$

$$\xi(t) = 3\frac{1+3z^{-1}}{1+\frac{1}{3}z^{-1}} e(t) \qquad \xi(t) \sim WN(0, 3^2 \cdot 3^2)$$

$\Longrightarrow \ b^* = \frac{1}{3} \qquad \lambda^{*2} = 81$

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exercise 1

$S: \ y(t) = e(t) + \frac{1}{3} e(t-1) \qquad e(t) \sim WN(0,1)$
$m: \ y(t) = -a y(t-1) + \eta(t-1) \qquad \eta(t) \sim WN(0,\lambda^2)$

$\hat{y}(t|t-1) = -a y(t-1)$

NOTE: we do not substitute the real $y(t)$

$$\varepsilon(t|t-1) = y(t) - \hat{y}(t|t-1) = (1 + a z^{-1}) y(t)$$
$$\mathbb{E}[\varepsilon(t|t-1)^2] = \mathbb{E}[(y(t) + a y(t-1))^2] =$$
$$= \mathbb{E}[y(t)^2] + a^2 \mathbb{E}[y(t-1)^2] + 2a\,\mathbb{E}[y(t)y(t-1)]$$
$$= \gamma_y(0) + a^2 \gamma_y(0) + 2a\,\gamma_y(1)$$

$$\frac{d}{da}\mathbb{E}[\varepsilon^2] = 2a\gamma_y(0) + 2\gamma_y(1) = 0 \implies a^* = -\frac{\overset{\text{from } \mathcal{S}}{\gamma_y(1)}}{\underset{\text{from } \mathcal{S}}{\gamma_y(0)}} = -\tfrac{1}{3} / \tfrac{10}{9} = \underbrace{-\tfrac{3}{10}}$$

We arrive at the same result

Model is stationary because $|a^*| < 1$

$$\lambda^* = \bar{J}(a^*) = \left(1 + \tfrac{9}{100}\right)\tfrac{10}{9} + 2\cdot\left(-\tfrac{3}{10}\right)\left(\tfrac{1}{3}\right) = \tfrac{81}{90} \sim 1$$

exercise 2

$\mathcal{S}: \; y(t) = \frac{1}{3}y(t-1) + v(t-1) + \eta(t) - \frac{1}{2}\eta(t-1)$ $\quad \left.\begin{array}{l}\eta(t) \sim WN(0,1)\\ v(t) \sim WN(0,1)\end{array}\right\}$ UNCORR.

$m: \; y(t) = -a\,y(t-1) + b\,v(t-1) + e(t)$ $\quad e(t) \sim WN(0,\lambda^2)$

$$\hat{y}(t|t-1) = \frac{F(z)}{C(z)}y(t) + \frac{B(z)}{C(z)}v(t) = \frac{-a}{1}y(t-1) + \frac{b}{1}v(t-1)$$

$$\begin{array}{r|l} 1 & 1 + az^{-1} \\ \hline -1 - az^{-1} & 1 \\ \hline / - az^{-1} & \end{array}$$

$$\varepsilon(t|t-1) = y(t) - \hat{y}(t|t-1) = y(t) + a\,y(t-1) - b\,v(t-1)$$

$$\mathbb{E}[\varepsilon^2] = \dots = \mathbb{E}[y(t)^2] + a^2\mathbb{E}[y(t-1)^2] + b^2\mathbb{E}[v(t-1)^2] + 2a\,\mathbb{E}[y(t)y(t-1)] - 2b\,\mathbb{E}[y(t)v(t-1)] - 2ab\,\mathbb{E}[y(t-1)v(t-1)] =$$

$$\bar{J}(\theta) = (1+a^2)\gamma_y(0) + b^2\gamma_v(0) + 2a\gamma_y(1) - 2b\,\mathbb{E}[y(t)v(t-1)] - \cancel{2ab\,\mathbb{E}[y(t-1)v(t-1)]} \to 0$$

$y = f(v(t-1), v(t-2), \dots)$

$$\begin{cases} \frac{\partial}{\partial a}\bar{J}(\theta) = \dots = 2a\gamma_y(0) + 2\gamma_y(0) = 0 \\ \frac{\partial}{\partial b}\bar{J}(\theta) = \dots = 2b\gamma_v(0) - 2\,\mathbb{E}[y(t)v(t-1)] = 0 \end{cases}$$

$$\to \begin{cases} a^* = -\gamma_y(1)/\gamma_y(0) = \dots = -\tfrac{7}{67} \\ b^* = \mathbb{E}[y(t)v(t-1)]/\gamma_v(0) = 1 \end{cases}$$

$$\mathbb{E}[y(t)v(t-1)] = \mathbb{E}\left[\left(\tfrac{1}{3}y(t-1) + v(t-1) + \eta(t) - \tfrac{1}{2}\eta(t-1)\right)v(t-1)\right] = \mathbb{E}[v(t-1)^2] = \gamma_v(0) = 1$$

Our predictor is stationary since $|a^*| < 1$.

$\lambda^* = \dots$

exercise 3

$\mathcal{S}: \; y(t) = e(t) + e(t-1) + e(t-2) \qquad e(t) \sim WN(0,1) \rightarrow MA(2)$

$m: \; y(t) = \eta(t) + \theta \eta(t-1) \qquad \rightarrow MA(1)$

Non canonical form for system. However, since our model is not of the same class, we don't really care.

$$\varepsilon(t) = \frac{A_m(z)}{C_m(z)} y(t) = \left(1 / 1 + \theta z^{-1} \right) \left(1 + z^{-1} + z^{-2} \right) e(t)$$

$$= -\theta \varepsilon(t-1) + e(t) + e(t-1) + e(t-2)$$

$$\mathbb{E}[\varepsilon(t)^2] = \theta^2 \mathbb{E}[\varepsilon(t-1)] + 3\mathbb{E}[e(t)] +$$
$$- 2\theta \mathbb{E}[\varepsilon(t-1)e(t)] - 2\theta \mathbb{E}[\varepsilon(t-1)e(t-1)] - 2\theta \mathbb{E}[\varepsilon(t-1)e(t-2)] +$$
$$+ \cancel{2\mathbb{E}[e(t)e(t-1)]} + \cancel{2\mathbb{E}[e(t)e(t-2)]} + \cancel{2\mathbb{E}[e(t-1)e(t-2)]}$$

$$\bar{J} = \theta^2 \bar{J} + 3 - \cancel{2\theta \mathbb{E}[\varepsilon(t-1)e(t)]} - 2\theta \left(\mathbb{E}[\varepsilon(t-1)e(t-1)] + \mathbb{E}[\varepsilon(t-1)e(t-2)] \right)$$

$$\mathbb{E}[\varepsilon(t-1)e(t-1)] = \mathbb{E}[(-\theta\varepsilon(t-2) + e(t-1) + e(t-2) + e(t-3))e(t-1)] = \dots = 1$$

$$\mathbb{E}[\varepsilon(t-1)e(t-2)] = \dots = -\theta + 1$$

$$= \theta^2 \bar{J} + 3 - 2\theta - \theta + 1 = 3 - 4\theta + 2\theta^2 / 1 - \theta^2$$

$$\frac{d}{d\theta} \bar{J} = -\frac{4 + 10\theta - 4\theta^2}{(1-\theta^2)^2} \implies \theta^* = 1/2$$

exercise 4

$\mathcal{S}: \; y(t) = -1/2 \, y(t-1) + e(t) \qquad e(t) \sim WN(0,1)$

$m: \; y(t) = -a \, y(t-2) + \eta(t) \qquad \eta(t) \sim WN(0, \lambda^2)$

$$\varepsilon(t) = \dots = y(t) - a\, y(t-2)$$

$$\mathbb{E}[\varepsilon(t)^2] = \mathbb{E}[y(t)^2] + a^2 \mathbb{E}[y(t-2)^2] - 2a\mathbb{E}[y(t)y(t-2)]$$

$$= (1 + a^2)\gamma_y(0) - 2a\gamma_y(2) = \bar{J}$$

from $\mathcal{S}$

$$\frac{d}{da}\bar{J} = 2a\gamma_y(0) - 2\gamma_y(2) = 0 \implies a^* = -\gamma_y(2) / \gamma_y(0) = \cdot 1/4$$

from $\mathcal{S}$

System is stationary since $|a^*| < 1$

exercise 5

$\mathcal{S}: y(t) = 3e(t) + e(t-2) \qquad e(t) \sim WN(0,1)$

$m: y(t) = a_1 y(t-1) + a_2 y(t-2) + \eta(t) \qquad \eta(t) \sim WN(0,\lambda^2)$

$$\varepsilon(t) = y(t) - \hat{y}(t|t-1) = \frac{A_m(z)}{C_m(z)} y(t) = \frac{1 - a_1 z^{-1} - a_2 z^{-2}}{1} y(t)$$

$$\mathbb{E}[\varepsilon(t)^2] = \dots =$$

29/03/23

exercise 1

$y(1) = 1 \qquad y(2) = 0 \qquad y(3) = -1 \qquad \mathcal{M}: y(t) = a\,y(t-1) + e(t) + a\,e(t-1)$

$e(t) \sim WN(0,\lambda^2)$

STEP 1: Compute the prediction form

$$y(t) = \frac{1 + az^{-1}}{1 - az^{-1}} \xi(t) \xrightarrow{\text{SIMP. FORM.}} \hat{y}(t|t-1) = \frac{C_m(z) - A_m(z)}{C_m(z)} y(t) = \dots =$$

$$= \frac{+2az^{-1}}{1 + az^{-1}} y(t)$$

$$= -a\,\hat{y}(t-1|t-2) + 2a\,y(t-1)$$

t	y(t)	$\hat{y}(t\|t-1)$
0	0	$\hat{y}(0\|-1) = 0$
1	1	$\hat{y}(1\|0) = -a\,\hat{y}(0\|-1) - 2a\,y(0) = 0$
2	0	$\hat{y}(2\|1) = -a\,\hat{y}(1\|0) - 2a\,y(1) = 2a$
3	-1	$\hat{y}(3\|2) = \dots = -2a^2$

$$\bar{J}_3 = \frac{1}{3} \sum_{i=1}^{3} \left(y(i) - y(i|i-1)\right)^2 =$$

$$= \frac{1}{3}\left[(1-0)^2 + (0-2a)^2 + (-1-2a^2)^2\right] =$$

$$= \frac{1}{3}\left[1 + \cancel{4a^2} + 1 + 4a^4 - \cancel{4a^2}\right] = \frac{1}{3}\left[2 + 4a^4\right] \Rightarrow \hat{a} = 0$$

exercise 2

$u(t) = -1 \qquad y(t) = \begin{cases} 1 & t \text{ odd} \\ -1 & t \text{ even} \end{cases}$

$$\mathcal{M}: \; y(t) = a\,y(t-1) + b\,u(t-1) + \xi(t) \qquad \xi(t) \sim WN(0, \lambda^2)$$
$$\mathcal{M}: \; \hat{y}(t|t-1) = a\,y(t-1) + b\,u(t-1)$$

$$\hat{J}_{15} = 1/15 \sum_{i=1}^{15} \left(y(i) - y(i|i-1)\right)^2$$

ARX => we can use least squares

$$\mathcal{M}: \; y(t) = \theta^T \varphi(t) + \xi(t) \qquad \varphi(t) = \begin{bmatrix} y(t-1) \\ u(t-1) \end{bmatrix}$$
$$\hat{\mathcal{M}}: \; \hat{y}(t|t-1) = \theta^T \varphi(t)$$

$$\Rightarrow \; \hat{\theta}_{15} = \left[\sum_1^{15} \varphi(i)\varphi(i)^T\right]^{-1} \cdot \sum_1^{15} \varphi(i)\,y(i) \quad \text{-> least square}$$

$$= \left[\sum_1^{15} \begin{bmatrix} y(i-1) \\ u(i-1) \end{bmatrix} \begin{bmatrix} y(i-1) & u(i-1) \end{bmatrix}\right]^{-1} \cdot \sum_1^{15} \begin{bmatrix} y(i-1) \\ u(i-1) \end{bmatrix} y(i)$$

$$= \begin{bmatrix} \sum_1^{15} y(i-1)^2 & \sum_1^{15} y(i-1)u(i-1) \\ \sum y(i-1)u(i-1) & \sum_1^{15} u(i-1)^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y(i-1)y(i) \\ \sum u(i-1)y(i) \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 1 \\ 1 & 15 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

exercise 3

$$\mathcal{M}: \; y(t) = 1/4\, y(t-1) - 1/a\, y(t-2) + e(t) \qquad e(t) \sim WN(0, \lambda^2)$$

$y(0) = 2 \quad y(1) = 0 \quad y(2) = -1$

STEP 1:

$$\hat{y}(t|t-1) = 1/4\, y(t-1) - 1/a\, y(t-2) \qquad a \neq 0$$

t	$y(t)$	$\hat{y}$
0	2	$\hat{y}(0\|-1) = 1/4\, y(-1) - 1/a\, y(-2) = 0$
1	0	$\hat{y}(1\|0) = 1/4\, y(0) - 1/a\, y(-1) = 1/2$
2	-1	$\hat{y}(2\|1) = 1/4\, y(1) - 1/a\, y(0) = 2/a$

$$J_3 = 1/3\left[(2-0)^2 + (0 - 1/2)^2 + (-1 - 2/a)^2\right] = \dots = 7/4 + 4/3\,\frac{a+1}{a^2}$$

$$\frac{d}{da} J_3 = 4/3 \; \frac{-a^2-2a}{a^4} = 0 \quad \longrightarrow \quad \hat{a} = -2$$

is $y(t) = 1/4 \; y(t-1) - 1/2 \; y(t-2) + e(t)$ canonical?

$$y(t) = \frac{1}{1 - 1/4 \, z^{-1} + 1/2 \, z^{-2}} \, e(t) \implies z^2 + 1/4 \, z + 1/2 = 0$$

$\hookrightarrow z_{1,2} = 1/8 \pm j \frac{\sqrt{31}}{8}$

$|z_{1,2}| = 1/2 < 1$